

Machine Learning

Regression Methods 2

FAST

DISCOVERING
THE FUTURE

Topics of previous lectures

- ✓ Ingredients of Machine Learning
- ✓ Classification Basics, Basic Linear Classifier
- ✓ K-Nearest Neighbours Classifier
- ✓ Naive Bayes Classifier
- ✓ Linear and Quadratic Discriminant Analysis
- ✓ Support Vector Machines (SVM)
- ✓ Decision Trees
- ✓ Ensemble Methods (Bagging, Weighted Voting, Stacking)
- ✓ Linear Regression

Topics of today's lecture

- Regularization
- Maximum Likelihood Estimation
- Logistic Regression
- Softmax Classifier
- Support Vector Regression (SVR)
- Regression Trees
- Ensemble Methods

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- Simpler models have less capacity to overfit the noise

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- Here λ is a regularisation parameter, which we can choose by cross-validation:
 - Higher λ means stronger regularization
 - If $\lambda = 0$ then we are back at OLS

Calculating the partial derivatives

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- Based on properties of matrix derivatives:

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- For ridge regression there exists a **closed-form solution** (i.e. can be explicitly calculated without numerical optimization):

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Con: all coefficients are still non-zero, even if the true model is very sparse
(very few non-zero coefficients)

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- No closed-form solution exists, must be optimised numerically

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- The parameter can be learned by using a hold-out validation dataset or K-fold cross-validation

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Lasso regression

- Pro:** if the true model is sparse then outperforms ridge regression and OLS
- Con:** if several features are highly correlated then tends to put high weight on an arbitrary one of those and zero to others

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- We can use the logistic sigmoid function on the predictions!

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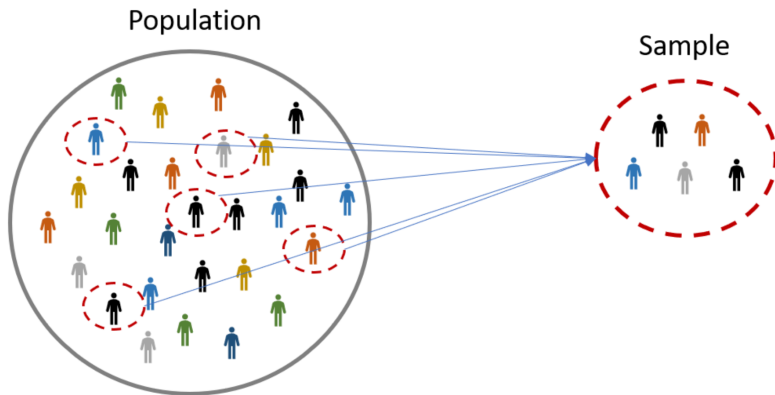
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- In Logistic Regression the parameters are usually fit by **Maximum Likelihood Estimation (MLE)** method

Motivation for MLE: Population vs Sample



Maximum Likelihood Estimation (MLE)

Definition (Likelihood function)

Let $f(x_1, \dots, x_n; \theta)$, $\theta \in \mathbb{R}^k$ be the joint PMF or PDF of random variables $X_1, \dots, X_n$ with sample values $x_1, \dots, x_n$. The **likelihood function** of the sample is given by

$$L(\theta; x_1, \dots, x_n) = L(\theta) = f(x_1, \dots, x_n; \theta)$$

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- continuous r.v. with density $f(x, \theta)$, then similarly

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Maximum Likelihood Estimation (MLE)

Definition (Maximum Likelihood Estimators)

The maximum likelihood estimators (MLEs) are those values of the parameters that maximize the likelihood function with respect to the parameter θ . That is,

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} L(\theta; x_1, \dots, x_n)$$

Maximum likelihood estimates give the parameter values for which the observed sample is **most likely** to have been generated.

Example

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In statistical terms, we call $\hat{\mu}$ and $\hat{\sigma}$ **point estimators** or **statistics**.

Definition (Point Estimator)

Let $x_1, x_2, \dots, x_n$ be a set of i.i.d. data points. A **point estimator** or **statistic** is any function of the data

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Since the data is drawn from a random process, any function of the data is random. Therefore $\hat{\theta}$ is a random variable.

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Definition (Bias)

The bias of an estimator is defined as

$$\text{bias}(\hat{\theta}_n) = \mathbb{E}(\hat{\theta}_n) - \theta,$$

where the expectation is taken over the data and θ is the true underlying value used to define the data-generating distribution.

Definition (Unbiased Estimator)

An estimator $\hat{\theta}_n$ is said to be **unbiased** if $bias(\hat{\theta}_n) = 0$, which implies that $\mathbb{E}(\hat{\theta}_n) = \theta$.

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Definition (Consistency)

An estimator is weak consistent, if $\hat{\theta}_n \xrightarrow{p} \theta \quad n \rightarrow \infty$ and strong consistent, if $\hat{\theta}_n \xrightarrow{a.s.} \theta \quad n \rightarrow \infty$.

Example

The MLE estimator of the Gaussian mean parameter is unbiased. Suppose $x_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$, $i = 1, \dots, n$, then

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The unbiased sample variance estimator is

$$\frac{1}{n-1} \sum_{i=1}^n (x_i - \hat{\mu}_n)^2.$$

MLE for Logistic Regression

- The likelihood function will be

$$L = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i},$$

where $p_i = P(y_i = \oplus | \mathbf{x}_i) = \sigma(\mathbf{w}^T \mathbf{x}_i)$

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- It is more convenient to work with the log likelihood function

$$\begin{aligned} \log(L) &= \sum_i \log(p_i^{y_i} (1 - p_i)^{1-y_i}) \\ &= \sum_i \left(\log(p_i^{y_i}) + \log((1 - p_i)^{1-y_i}) \right) = \\ &= \sum_i \left(y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right) \end{aligned}$$

Does this look familiar?

MLE for Logistic Regression

Now that we have the log likelihood function, we can compute the partial derivative w.r.t. $\mathbf{w}$

$$\frac{\partial \log(L(\mathbf{w}))}{\partial \mathbf{w}} = \frac{\partial}{\partial \mathbf{w}} \sum_i (y_i \log(\sigma(\mathbf{w}^T \mathbf{x}_i)) + (1 - y_i) \log(1 - \sigma(\mathbf{w}^T \mathbf{x}_i))) =$$

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- The loss function that logistic regression minimizes is known as **cross entropy** (also known as log-loss or binomial deviance)

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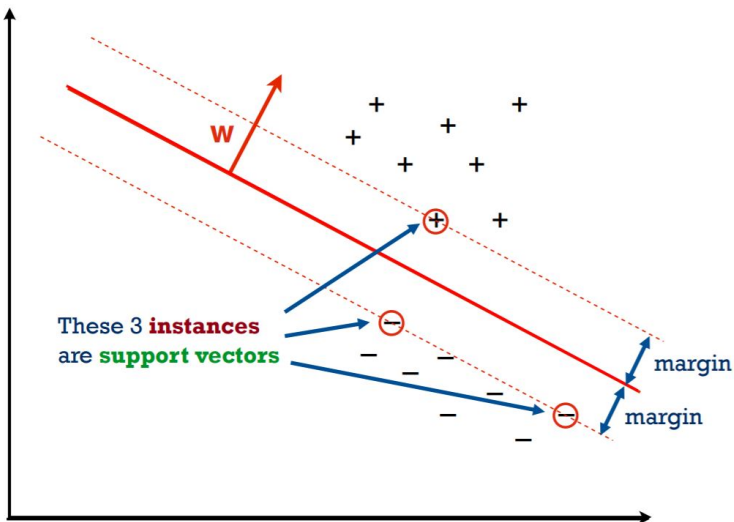
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- The problem can be solved with gradient descent.

Support Vector Machine (SVM) for Classification



Hard-margin SVM for classification

$$\mathbf{w}^*, t^* = \operatorname{argmin}_{\mathbf{w}, t} \frac{1}{2} \|\mathbf{w}\|^2$$

subject to $y_i(\mathbf{w} \cdot \mathbf{x}_i - t) \geq 1, \quad y_i \in \{-1, 1\}, \quad i = 1, \dots, n$

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We can solve the equivalent dual problem

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From the result we can calculate:

$$\mathbf{w}^* = \sum_{i=1}^n \alpha_i^* y_i \mathbf{x}_i \quad t^* = \mathbf{w}^* \cdot \mathbf{x}_i - y_i,$$

where $\mathbf{x}_i$ is a support vector and α_i is its weight.

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$$\mathbf{w}^*, t^*, \xi_i^* = \operatorname{argmin}_{\mathbf{w}, t, \xi_i} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i$$

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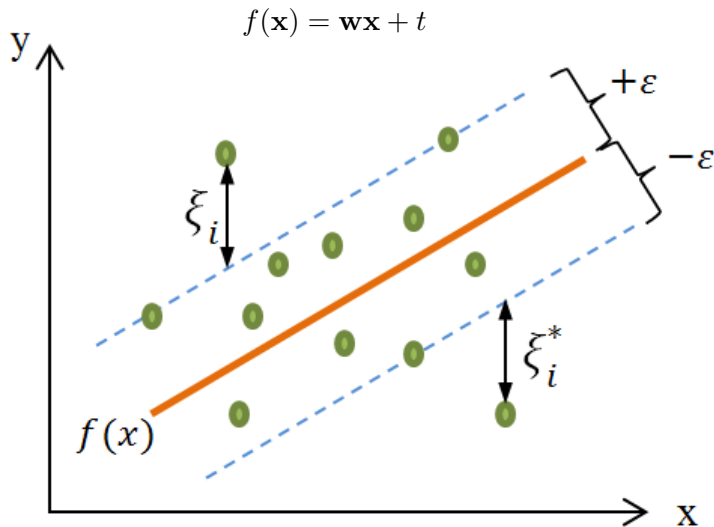
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SVM for Regression



Support Vector Regression

$$\min_{\mathbf{w}, t, \xi_i, \xi'_i} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n (\xi_i + \xi'_i)$$

$$\text{subject to } y_i - \mathbf{w} \cdot \mathbf{x}_i - t \leq \epsilon + \xi_i, \quad \xi_i \geq 0$$

$$\mathbf{w} \cdot \mathbf{x}_i + t - y_i \leq \epsilon + \xi'_i, \quad \xi'_i \geq 0, \quad i = 1, \dots, n$$

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We can solve the equivalent dual problem

$$\max_{\boldsymbol{\alpha}; \boldsymbol{\alpha}'} -\frac{1}{2}(\boldsymbol{\alpha} - \boldsymbol{\alpha}')^T Q(\boldsymbol{\alpha} - \boldsymbol{\alpha}') - \epsilon e^T(\boldsymbol{\alpha} + \boldsymbol{\alpha}') + y^T(\boldsymbol{\alpha} - \boldsymbol{\alpha}')$$

$$\text{subject to } 0 \leq \alpha_i, \alpha'_i \leq C, \quad i = 1, \dots, n \text{ and } e^T(\boldsymbol{\alpha} - \boldsymbol{\alpha}') = 0$$

where $Q = \mathbf{xx}^T$ and $e = (1, \dots, 1)^T$

Support Vector Regression

$$\min_{\mathbf{w}, t, \xi_i, \xi'_i} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n (\xi_i + \xi'_i)$$

$$\text{subject to } y_i - \mathbf{w} \cdot \mathbf{x}_i - t \leq \epsilon + \xi_i, \quad \xi_i \geq 0$$

$$\mathbf{w} \cdot \mathbf{x}_i + t - y_i \leq \epsilon + \xi'_i, \quad \xi'_i \geq 0, \quad i = 1, \dots, n$$

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$$\max_{\alpha; \alpha'} -\frac{1}{2} (\alpha - \alpha')^T Q (\alpha - \alpha') - \epsilon e^T (\alpha + \alpha') + y^T (\alpha - \alpha')$$

$$\text{subject to } 0 \leq \alpha_i, \alpha'_i \leq C, \quad i = 1, \dots, n \text{ and } e^T (\alpha - \alpha') = 0$$

where $Q = \mathbf{xx}^T$ and $e = (1, \dots, 1)^T$

From this we can calculate the prediction: $\hat{f}(\mathbf{x}) = \mathbf{w}^* \cdot \mathbf{x} + t$, with

$$\mathbf{w}^* = \sum_{i=1}^n (\alpha_i - \alpha'_i) \mathbf{x}_i, \quad t^* = y_i - \epsilon - \mathbf{w}^* \cdot \mathbf{x}_i$$

Support Vector Regression

Can we apply kernels in this case?

$$\min_{\mathbf{w}, t, \xi_i, \xi'_i} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n (\xi_i + \xi'_i)$$

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- Fitting (soft-margin kernel SVR):

$$\max_{\alpha; \alpha'} -\frac{1}{2}(\alpha - \alpha')^T Q(\alpha - \alpha') + \epsilon e^T(\alpha + \alpha') - y^T(\alpha - \alpha')$$

subject to $0 \leq \alpha_i, \alpha'_i \leq C, \quad i = 1, \dots, n$ and $e^T(\alpha - \alpha') = 0$

where $Q = \kappa(\mathbf{x}_i, \mathbf{x}_j)$ and $e = (1, \dots, 1)^T$

$$t^* = y_j - \epsilon - \sum_{i=1}^n (\alpha_i - \alpha'_i) \kappa(\mathbf{x}_i, \mathbf{x}_j),$$

where $\mathbf{x}_j$ is a support vector of target value y_j .

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- Prediction:

$$\hat{f}(\mathbf{x}) = \sum_{i=1}^n (\alpha_i - \alpha'_i) \kappa(\mathbf{x}_i, \mathbf{x}) + t^*$$

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- A split is considered good if it maximizes the impurity gain:

$$Imp(D) - Imp(\{D_1, \dots, D_l\})$$

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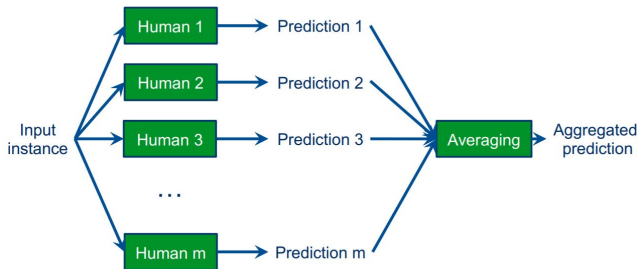
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- In case of regression we take $Imp = Var$ and we base our decision on **variance reduction**
- Each leaf returns the average target value of the segment

- In case of **bagging**, we take the **average prediction** across the individual models

Ensemble Methods

- In case of **bagging**, we take the **average prediction** across the individual models



- In case of **Random Forest**, we grow a forest of **regression trees** and take the average prediction across the trees. The number of features in each tree is usually taken as $F = \frac{P}{3}$, where P is the number of total features in the dataset

What have we learned today?

- ✓ Regularization
- ✓ Maximum Likelihood Estimation
- ✓ Logistic Regression
- ✓ Softmax Classifier
- ✓ Support Vector Regression (SVR)
- ✓ Regression Trees
- ✓ Ensemble Methods